Krittin Chaowakarn

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EDUCATION

Master of Science in Open Informatics 09/2026–Present
Czech Technical University in Prague (CTU) Prague, Czech Republic
Specialization: Computer Vision

Bachelor of Engineering in Electrical Engineering 08/2021–11/2025
Sirindhorn International Institute of Technology, Thammasat University Pathum Thani, Thailand

- GPA: 3.87/4.00 – **First-Class Honors**

TUMexchange (Exchange Semester) 04/2025–09/2025
Technical University of Munich Munich, Germany

- Thesis: *"Real-Time Object Detection for Autonomous Driving: An Empirical Study in a Small-Scale Urban Environment"*, supervised by Prof. Andreas Herkersdorf

PREPRINTS

K. Chaowakarn, P. Sangwongngam, N. H. H. Aung, and C. Charoenlarpnopparut, "*NV3D: Leveraging Spatial Shape Through Normal Vector-based 3D Object Detection*," *arXiv preprint* arXiv:2510.11632, 2025. [Online]. Available: `https://arxiv.org/abs/2510.11632`

RESEARCH EXPERIENCE

Researcher (0.5 FTE) 10/2026–Present
Czech Institute of Informatics, Robotics and Cybernetics (CIIRC), CTU Prague, Czech Republic
Research Group: Intelligent Machine Perception (IMPACT) Team

- Developing simulation environments for robot manipulation

Bachelor's Thesis Student 04/2025–09/2025
Technical University of Munich Munich, Germany
Advisor: Prof. Andreas Herkersdorf

- Collected and curated an image dataset from the lab environment using k-means clustering on L2-normalized embeddings (unit hypersphere representation), reducing ∼1,000 raw images to 119 distinct samples
- Conducted an empirical study on YOLO-based real-time object detection in a small-scale urban environment using low-power NVIDIA Jetson Nano, evaluating latency, throughput, and hardware utilization

Undergraduate Researcher / Research Intern 06/2023–04/2025
National Electronics and Computer Technology Center Pathum Thani, Thailand
Advisors: Dr. Paramin Sangwongnam and Assoc. Prof. Chalie Charoenlarpnopparut

- Conducted research on LiDAR-based 3D multi-object detection for autonomous vehicles and proposed two normal-vector-guided sampling strategies that reduced redundant points by up to 55%
- Accelerated normal vector computation by 6.67× through CUDA C++ parallelization, reducing runtime from 0.10 s (PyTorch) to 0.015 s

Technical Collaborator 01/2025–03/2025
Sirindhorn International Institute of Technology, Thammasat University Pathum Thani, Thailand
Advisor: Assoc. Prof. Itthisek Nilkhamhang

- Contributed to multi-robot formation research using Yahboom robots and ROS 2 simulations
- Deployed a vision-based target tracking system integrating YOLO (detection), DeepSORT (tracking), and face recognition (identity verification) for human-following robots

Research Intern 06/2022–07/2022
National Electronics and Computer Technology Center Pathum Thani, Thailand
Advisor: Dr. Paramin Sangwongnam

- Performed a deep feedforward neural network using TensorFlow for mmWave beam and blockage prediction utilizing Sub-6 GHz signals

TEACHING/ACADEMIC SERVICE

Teaching Assistant 01/2023–Present
Sirindhorn International Institute of Technology, Thammasat University Pathum Thani, Thailand

- Graduate-level course grading: Computational Mathematics: Probability (M.Eng. in AI and IoT, SkillLane Online Platform)
- Laboratories & Lectures: Control Systems, Electronics, Digital Circuits, Electromagnetics, Linear Algebra and Optimization Method, Basic Electrical Engineering

Research Mentor 03/2026–Present
National Electronics and Computer Technology Center Pathum Thani, Thailand

- Mentored high school interns in developing a self-supervised method for Thai temple crack detection using a PointNet++-based encoder

WORK EXPERIENCE

AI Engineer Intern 06/2024–07/2024
BOTNOI Group Bangkok, Thailand

- Improved Thai SNOMED-CT medical vocabulary using a machine translation model, achieving over 80% reduction in translation ambiguities (70 out of 86 issues solved from a dataset of 385)
- Developed a greedy, embedding-based alignment algorithm for English-Thai (unsegmented) text to build genre-specific MT datasets, cutting manual matching time by 80% (50 to 10 hours)

HONORS AND AWARDS

Young Scientist and Technologist Program 08/2021–05/2025

- Full scholarship from the National Science and Technology Development Agency; 1 of 4 awardees ($\sim$0.25%)

Top 100 Team – KPIT Sparkle 2024 04/2024

- Selected among the top 100 teams ($\sim$0.45%) worldwide in KPIT Sparkle 2024, a global student innovation contest focused on vehicle technologies

SKILLS

Programming: Python (PyTorch, JAX), C/C++, Linux (CLI, Bash), Git
Languages: English (professional proficiency); Thai (native); German (basic)